

OPERATIONS RESEARCH AND SYSTEMS ANALYSIS

Simulation in MS Excel, simulation using ProModel

Task 1

- Use MS Excel and Monte Carlo simulation to estimate frequencies of number roll on dice (D6).
- ▣ Simulate 10, 100, 1000, 10000 rolls. Interpret the differences.

MS EXCEL:

Uniform distribution pseudorandom number

... =RAND()

Uniform distribution pseudorandom number from a to b

... =RAND()*($b-a$) + a

Integer pseudorandom number from a to b

... =RANDBETWEEN(a,b)

To count number of values in area

... =COUNTIF(area,value)

Task 2

- Use ProModel to solve the example from 4th lecture:
 - *Highway weighting machine*
 - *Kendall notation: $M/M/1$*
 - *Arrivals: one customer per 4 minutes*
 - *Service time: 18 customers can be served per 1 hour*
 - Calculate the probability that the customer will not wait.
 - What is average length of queue?
 - What is the average time spent in system?

Thank you for your attention.

